PRODUCT(IP/IC) DEVELOPMENT PROCESS

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Quality management Div.

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Rev. no	Rev. date	Simple description of changed contents	Author	Post
1.0	2022.09.22	Newly established.	Mark.Kim	QM

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1. Purpose and scope

This document defines the standard development process for Intellectual Property (IP) and Integrated Circuit (IC) products. The objective is to establish a consistent, high-quality, and efficient workflow across all projects to improve predictability, reduce development risk, and ensure reusability and compliance with design requirements.

2. Apply scope

This standard applies to all IP and IC development activities performed within the company, including digital, analog, and mixed-signal designs. It covers the entire lifecycle from specification through tape-out and post-silicon validation.

3. Definition of terms

3.1 IP (Intellectual property)

IP is a design database in a form that can be reused as design data whose process is completely defined and has been placed and connected

3.2 IC (Integrated Circuit)

A complete chip that includes multiple IPs, interconnects, and top-level integration.

3.3 RTL (Register Transfer Level)

A level of abstraction used to describe the digital circuit's functionality.

3.4 HDL (Hardware Description Language)

A programming language used to describe the behavior and structure of electronic circuits.

3.5 DV (Design Verification)

The process of checking whether the design meets its functional and performance specifications. This includes simulation, coverage analysis, assertion checks, formal verification, and regression testing.

3.6 Test bench

A simulation environment that includes stimulus generators, monitors, checkers, and scoreboards used to verify the correctness of the DUT(Design under test). May be written using UVM or other verification methodologies.

3.7 Coverage

A measure of how thoroughly the design has been verified. Types include code coverage(line, toggle, FSM, condition) and functional coverage(custom metrices based on the specification)

3.8 Linting

Static code analysis that checks for coding style, syntax errors, design rule violations, and potential functional issues in RTL code.

3.9 CDC(Clock Domain Crossing)

A situation where signals pass between different clock domains. CDC checks ensure that these crossings are properly synchronized to avoid metastability and data loss.

3.10 DFT(Design for Testability)

Techniques embedded into the design to make post-silicon testing easier and more effective, such as scan chains, BIST(Built in self test), and boundary scan.

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3.11 Synthesis

The process of translating RTL code into a gate-level netlist that can be used for physical implementation. It includes optimization for area, timing, and power.

3.12 STA(Static Timing Analysis)

A method for verifying the timing performance of a digital circuit without simulating every possible input combination, ensures that timing constraints are met under all conditions.

3.13 PPA(Power, Performance, Area)

The key design metrics used to evaluate the efficiency and suitability of an IP or IC.

Power: Total power consumption(dynamic and static)

Performance: Speed or throughput of the design.

Area: Silicon area occupied on the die.

3.14 Sign-off

A formal process where a phase or deliverable is reviewed and approved before moving the next state. Example: design sign-off, verification sign-off, synthesis sign-off

3.15 Release

The act of finalizing and distributing an IP version with all supporting files and documentation. The release must pass QA checks and follow version control protocols.

3.16 Integration

The process of incorporating the IP into a higher-level system, ensuring correct connectivity, configuration, and functionality with surrounding logic.

4. Responsibility and authority

4.1 CEO

1) Ensure company-wide commitment to quality and reliability

Promote a corporate culture that prioritizes project quality and long-term reliability

Ensure all departments (Design, Quality, Manufacturing, etc.) align with quality objectives.

2) Approve and allocate resources for qualification activities

Ensure sufficient budget, personnel, and infrastructure (e.g. lab equipment, test time) are allocated to perform comprehensive qualification and reliability test.

3) Define and endorse quality and reliability policies

Approve internal quality management policies that include qualification standards, failure criteria, and risk management procedures.

Support adoption of international standard (e.g. ISO9001, IATF16949, ACE-Q100) across the organization.

4) Make business-level decisions based on qualification outcomes

Make go/no-go decisions for product release based on qualification results and associated risks.

Approve escalation plans or mitigation strategies in case of qualification failure.

5) Represent the company in customer and regulatory matters

Serve as the final point of contact for critical customer communications regarding product quality and reliability.

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Take responsibility for regulatory compliance and customer audits if escalated.

6) Foster cross-functional accountability

Ensure that all departments collaborate effectively during the qualification process.

Hold executive staff accountable for fulfilling their roles in design, process, quality, and production readiness.

4.2 Design department (Analog solution, Digital solution, Smart solution)

1) Design compliance with project specification

Ensure that circuit and layout design meet the electrical and physical performance requirements as defined by customer and internal product specification

2) Design for reliability

Incorporate considerations for reliability factors such as high temperature, humidity, and electrical stress in to the design.

Include protection against ESD, Latch-up, aging and other reliability risks

3) Provide data for qualification support

Deliver necessary design data for qualification, including corner simulations and worst-case analysis results.

Design test chips or test vehicles required for reliability evaluations.

4) Support failure analysis

Provide technical support form the design perspective when failures occur during qualification.

Propose corrective action or design improvements as needed.

5) Reflect qualification results in design

Review and implement necessary design changes based on qualification outcome (e.g. margin enhancement, rule updates)

6) Documentation and design history management

Prepare and maintain reliability-related design documents such as design guidelines, DRC/LVS report, and design review records.

4.3 Quality management department

1) Establish and manage qualification plans

Define the qualification strategy, including applicable standard (e.g. JEDEC, AEC Q100) test conditions, sample size and criteria.

Coordinate with design, process and reliability teams to ensure alignment with customers and internal requirements.

2) Monitor and control the qualification process

Ensure that qualification tests are executed properly according to the defined plan and schedule.

Track progress, collect test data, and maintain traceability throughout the qualification cycle.

3) Interface with customers on quality requirements.

Communicate with customers regarding qualification expectations, report formats, schedules, and results.

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Address any customer feedback or concerns related to qualification and reliability.

4) Review and approve qualification results.

Analyze test data and determine whether the results meet acceptance criteria.

Provide formal approval (or rejection) of the qualification outcome, including release to production if successful.

5) Drive corrective and preventive actions

When failures occur, lead root cause analysis and corrective action processes across departments.

Ensure that lessons learned are documented and integrated into design, process or test improvements.

6) Maintain quality and reliability documentation

Create and manage qualification reports, failure analysis summaries and control plan.

Ensure documentation complies with internal (Quality management system) and customer audits.

7) Ensure compliance with industry standards

Enforce adherence to relevant standards such as ISO9001, IATF 16949, JEDEC, and AEC-Q series.

Support internal and external audits related to qualification and reliability processes.

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5. Process mapping

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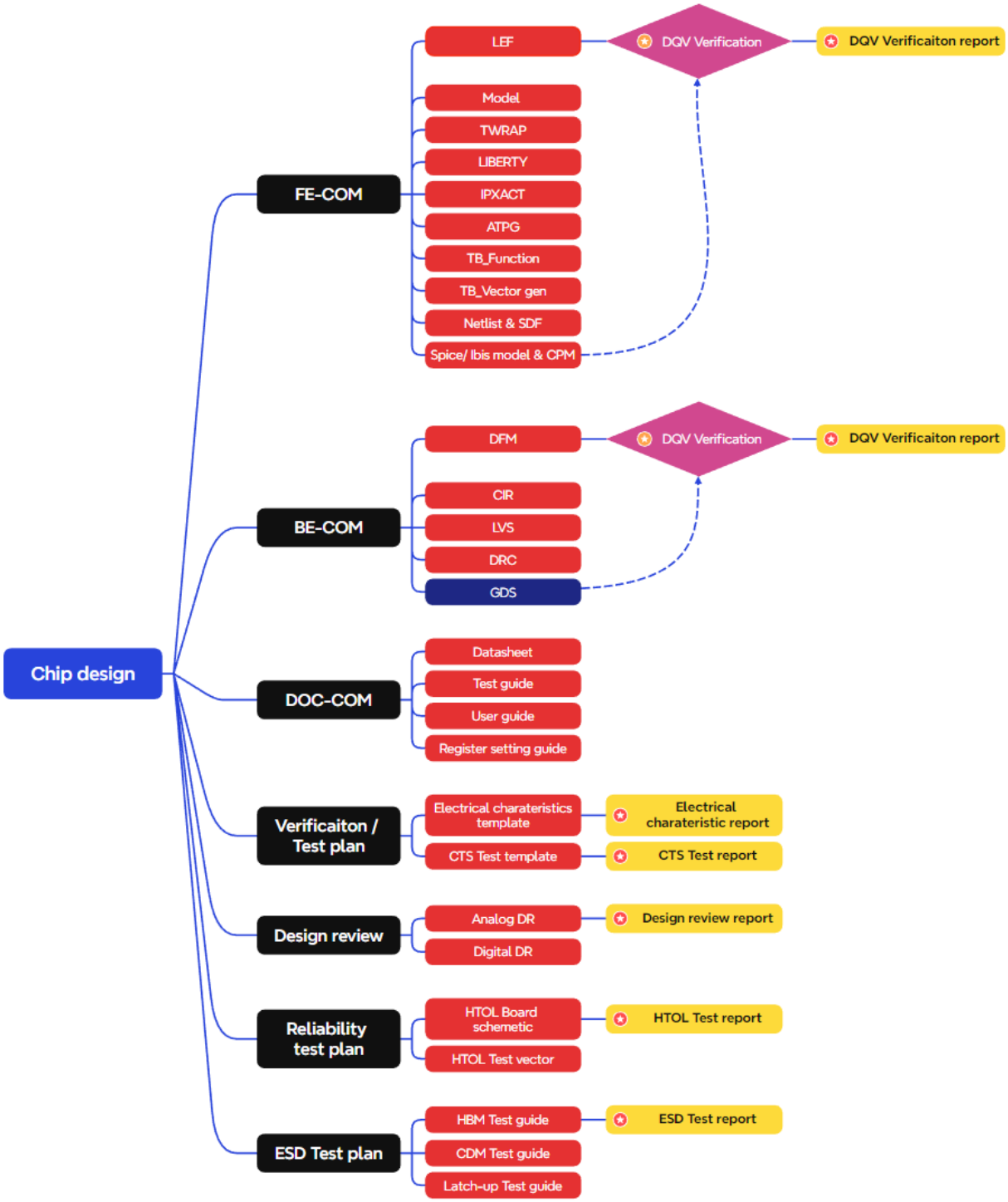
6. Work flow description



6.1 Design work flow

Development lifecycle

- **Planning phase**
  - Define IP specifications (functional, performance, interface, power, etc.).
  - Review market and customer requirements.
  - Identify dependencies, risks, and target process technology.
  - Allocate resources and schedule the development timeline.
  - Document the IP Specification Sheet.
- **Design Phase**
  - Architecture Definition
    - Define the internal structure of the IP based on the functional specification.
    - For digital IPs, this includes data path, control logic, FSMs, and interfaces.
    - For analog IPs, this includes high-level block diagram (e.g., bias generator, amplifier stages, charge pumps), signal flow, noise/power/speed trade-offs.
  - RTL Design (Digital IP)
    - Implement RTL code using Verilog/SystemVerilog/VHDL.
    - Perform code linting and apply coding style guides
    - Simulate basic functionality at the block level.
    - Conduct internal design reviews to validate logic correctness and architecture.
  - Analog Circuit Design (Analog IP)
    - Design analog blocks at the transistor level using industry-standard tools (e.g., Cadence Virtuoso).
    - Size transistors considering performance metrics such as gain, bandwidth, linearity, noise, and power consumption.
    - Run schematic simulations (e.g., DC, AC, transient, noise, corner, Monte Carlo) to validate functionality.
    - Perform layout-aware simulation or post-layout extraction if applicable.
  - Mixed-Signal Considerations (if applicable)
    - For mixed-signal IPs (e.g., SerDes, ADC/DAC), clearly define the analog-digital interface (e.g., boundary of behavioral models and digital control logic).
    - Develop Verilog-AMS or SystemVerilog behavioral models to support early verification and top-level simulation.
    - Ensure proper synchronization and isolation techniques between analog and digital domains (e.g., level shifters, ESD protection, isolation rings).
  - Layout Planning (Analog & Mixed-Signal)
    - Begin early layout planning to estimate area and validate floorplan feasibility.
    - Ensure layout complies with analog-specific design rules (matching, shielding, symmetry, guard rings, etc.).
    - Coordinate with physical design team for integration constraints and floorplan placement.
  - Review
    - Conduct formal design reviews involving both digital and analog designers.
    - Review schematics, simulation results, timing/power estimates, and design assumptions.
    - Identify risk areas, corner case behaviors, and limitations in the design.
- **Verificaiton Phase**
  - Implement testbenches and verification components.
  - Run simulation and regression tests.
  - Track functional coverage and verify spec completeness.
  - Conduct verification review and sign-off.
- **Integration & Validation**
  - Perform integration tests with other IPs or top-level systems.
  - Ensure compatibility and proper interface behavior.
  - Run SoC-level simulations if applicable.
  - Perform gate-level simulations or emulation (optional).
- **Physical Implementation (if applicable)**
  - Run synthesis and generate gate-level netlist.
  - Check timing (STA), area, and power reports.
  - Run DFT insertion and formal equivalence checks.
  - Review floorplan compatibility if part of a hard IP.
- **Documentation**
  - Datasheet / Test guide / User guide / Register seeting guide
- **Review & Sign-off process**
  - Design reivew
- **Change management**
  - All changes must follow a documented Change Request (CR) process.
  - CRs must include description, rationale, and potential impacts.
  - All modifications must be re-verified and re-reviewed.
- **Package assembly**
  - Package netlist / Substate PCB Lauout / Package assembly
- **Silicon validation**
  - Electrical charateristic / ESD test / Device reliability test / CTS Test

6.2 Verification work flow

Phase #1 (DQV Verification)

1) Once the design is completed and the database (DB) is ready, the design department requests DQV verification from the QA department.
2) The QA department conducts the DQV verification on the requested DB and then performs a consistency evaluation of the DB.

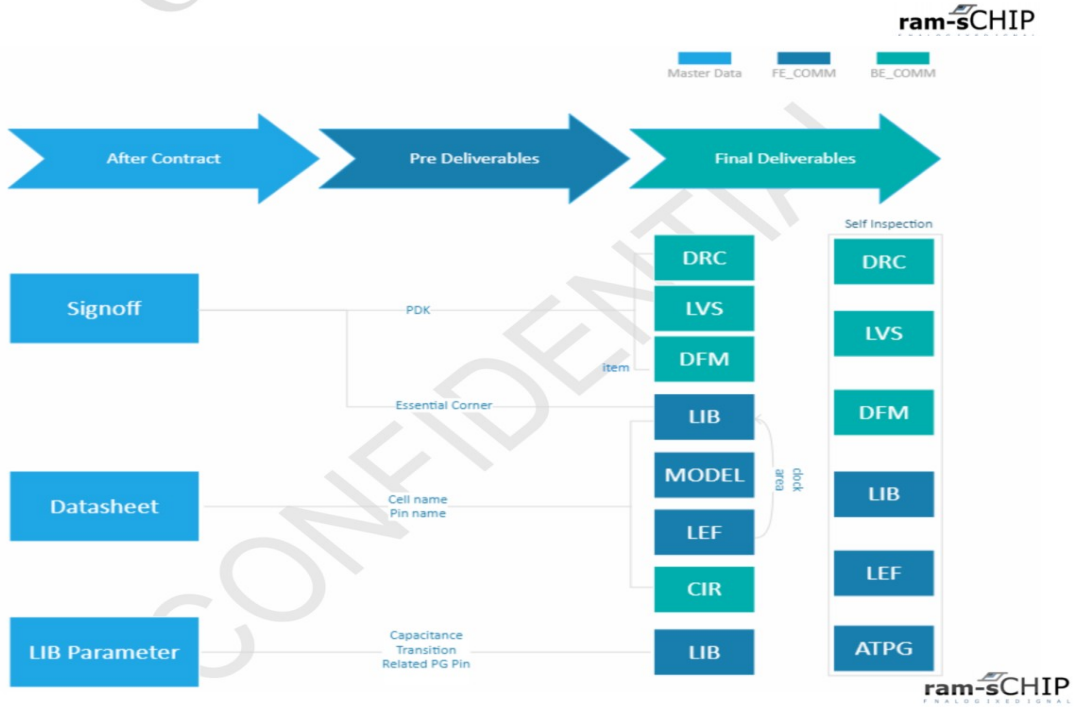
NOTE:

DQV (Design Quality Verification) is an in-house script tool developed by Ramschip to evaluate the consistency of the database (DB). The tool checks pin name matching between various

DQV Rule Decks

Total: 24 rules **F: FE_COMMON, B: BE_COMMON D: DOC_COMMON**

Rule #	F	B	D	Description	Verification method
1	V	V	V	File name and presence check (Auto load only)	Internal naming convention vs All files in deliverables
2	V	V	V	Cell name and presence check	Internal naming convention vs All files in deliverables
3	V	V	V	Pin name check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
4	V	V	V	Pin direction check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
5	V	V	V	Pin bit width check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
6	V			Cell area check	LEF vs LIBERTY
7	V			LEF Cell Consistency	Comparing within LEF files
8	V		V	P/G Pin Check	Comparing within LIB files (P/G Pin vs Voltage Map)
9	V			Liberty description (PVT) check	Comparing within LIB files (Library vs default_operating_conditions)
10	V			Essential Corner check	Signoff vs LIBERTY
11	V			Clock check	Comparing the clock pins (in the LEF file vs in the LIB files)
12	V			No timing arc	Comparing within LIB files
13	V			Timing arc with numbers	Comparing within LIB files
14	V			Intra-Liberty Related P/G Pin check	Voltage map vs LIB files
15	V			Related Pin	Comparing within LIB files
16	V			Max Capacitance check	Specified Value vs LIB files
17				Self-Capacitance check	Specified Value vs LIB files
18	V			Max Transition	Specified Value vs LIB files
19		V		PDK Version check	Signoff VS (LVS, DRC, DFM)
20	V			Test Coverage	ATPG Files
21		V	V	DRC Signoff Check	Signoff VS DRC (sourceme. *)
22		V	V	DFM Item check	Signoff VS DFM
23		V	V	LVS, DFM Error Log check	Within LVS, DFM Report
24		V	V	Bump Size Check	Customer Requirement (Orderform) VS DRC (sourceme.*)



deliverables and verifies simulation results.

Manual Link: [DQV User Manual V1.01.pdf]

Phase #2 (Electrical characteristics evaluation)

1) After the test chip is fabricated (fab-out) based on the designed database, it proceeds to package assembly.

Silicon Proven check list

1.1. General information

IP	RS_DCPHY6310_LN04LPP_M4S4	PDK Ver.	S00	Evaluation engineer	DS - Ian/ Brett	Final measurement date	
Sample package	144FCBGA	Corner sample	SS	QA Check engineer	QM - Ronnie	Final checking date	

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1.2. Silicon proven items & result

Clasification	Test items	Probe point	Vector	Symbol	Power Condition	Judgement criteria min	typ.	max.	unit	Measurement 2st Build Sample SS #1 -40 ℃	RT	125℃	2st Build Sample SS #2 -40 ℃	RT	125℃	2st Build Sample SS #11 -40 ℃	RT	125℃	CT MIN	MAX	RT MIN	MAX	HT MIN	MAX	Judgement (Pass/Fail) Sample #1	Sample #2	Sample #3
Scan Test	SCAN_TEST	M_SCAN_OUT_C0/1/2/3_TM[X:0] S_SCAN_OUT_C0/1/2/3_TM[X:0]	-	ALL and SCAN	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
IDS Current	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.85V) M_VDD12 (1.2V)	-		60	mA	0.9	1.8	9.7	0.8	1.4	7.8	0.8	1.6	10.1	0.80	0.90	1.40	1.80	7.80	10.10	On going	On going	
		M_VDD12		IDS_M_VDD12		-		10		0	0.002	0.002	0.001	0.001	0.002	0.001	0.001	0.002	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40		0.3	0.9	4.9	0.4	0.9	4.1	0.4	0.9	5.2	0.40	0.50	0.90	0.90	4.10	5.20	On going	On going	
		M_VDD12		IDS_M_VDD12		-		10		0	0.002	0.002	0.001	0.001	0.002	0	0.001	0.002	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA	0.1	0.1	0.35	0.1	0.13	0.42	0.11	0.11	0.37	0.10	0.11	0.10	0.13	0.35	0.42	On going	On going	
	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.7125V) M_VDD12 (1.14V)	-		60	mA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.8925V) M_VDD12 (1.26V)	-		60	mA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Master Core IP Unidirectional LP Mode Test	Master clock lane LP-TX function	M_DPCLK M_DNCLK M_TEST_OUT_C[1:0]	dphy_master_clock_lptx	LPTX_MC_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS		PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0.001	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MC_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.083	1.12	1.11	1.12	1.12	1.09	1.12	1.08	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Master data lane LP-TX function	M_DPDATA0/1/2/3 M_DNDATA0/1/2/3 M_TEST_OUT_0/1/2/3[1:0]	dphy_master_data_lptx	LPTX_MD_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0.001	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MD_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.084	1.12	1.11	1.12	1.12	1.09	1.12	1.08	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Master data lane LP-TX function(C-PHY)	M_DNDATA0_A0 M_DPDATA0_B0 M_DNDATA1_C0 M_DPDATA1_A1 M_DNCLK_B1 M_DPCLK_C1	cphy_master_data_lptx	LPTX_MD_CPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0	0	0.001	0	0	0.001	0	0	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MD_CPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.092	1.12	1.11	1.12	1.12	1.09	1.12	1.09	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0	0	0	0	0	0	0	0	0	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Master Core IP Bidirectional LP Mode Test	Mater data lane LP-RX function(D-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_DPHY	Condition #1 MS_VDD0P85 (0.8925V) M_VDD12 (1.26V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(C-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	D-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	C-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(D-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_DPHY	Condition #2 MS_VDD0P85 (0.7125V) M_VDD12 (1.14V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(C-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	D-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	C-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Slave Core IP Uni-direction LP Mode Test	Slave clock lane LP-RX function(D-PHY)	S_TEST_OUT_C[1:0]	dphy_slave_clock_lprx	SC_LPRX_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(D-PHY)	S_TEST_OUT_0/1/2/3[1:0]	dphy_slave_data_lprx	SD_LPRX_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(C-PHY)	S_TEST_OUT_0/1/2[1:0]	cphy_slave_data_lprx	SD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave clock lane LP-RX function(D-PHY)	S_TEST_OUT_C[1:0]	dphy_slave_clock_lprx	SC_LPRX_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(D-PHY)	S_TEST_OUT_0/1/2/3[1:0]	dphy_slave_data_lprx	SD_LPRX_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(C-PHY)	S_TEST_OUT_0/1/2[1:0]	cphy_slave_data_lprx	SD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
PLL Frequency	PLL 6.5GHz (DPHY)	PLL_TEST_DIV	dphy_ext(int)_lb_6500M.v	PLL_FREQ_DPHY_6p5G	Condition #1 MS_VDD0P85 (0.8925V)	#####	#####	######	MHz	101.280	101.360	101.309	101.280	#####	#####	#####	######	######	101.28	101.38	101.30	101.36	101.19	101.38	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.8075V)					101.376	101.299	101.306	101.312	#####	#####	#####	######	######	101.31	101.38	101.30	101.38	101.30	101.37	On going	On going	On going
	PLL 2.5GHz (DPHY)		dphy_ext(int)_lb_2500M.v	PLL_FREQ_DPHY_2p5G	Condition #1 MS_VDD0P85 (0.8925V)	38.770	38.965	39.160		38.944	38.928	38.922	39.072	38.922	39.107	39.072	39.060	39.066	39.07	39.07	38.92	39.06	38.92	39.11	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.7125V)					38.944	38.915	38.930	38.994	38.954	39.062	39.072	39.056	39.069	38.94	39.07	38.92	39.06	38.93	39.07	On going	On going	On going

2) QA engineer proceeds the electrical characteristics (ex/ standby current, dynamic current, PLL, Function loop-back test, etc.) based on the test plan or test checklist.
3) If the measured results fall outside the acceptance criteria, The QA engineer notifies the design department, and they conduct a root cause analysis.
4) If the items outside the acceptance criteria can be de-bugged, the engineers should perform the debugging and then request re-qualification from the QA engineer.
5) QA engineer performs re-qualification if the troubleshooting is deemed reasonable.

Phase #3 (compliance Test Suite)

1) Once the electrical characteristics evaluation is complete, QA / Design engineers should perform compliance test (CTS) to ensure that the designed IP meet the required standard.
2) CTS test should be performed based on the items and criteria defined in the CTS standard.
3) The linked CTS report follows Ramschip's official report format.

[CTS REPORT_rs_dcphy6310a_ln04lpp.pdf]

Phase #4 (ESD Test / Device life Test)

1) The QA engineer should prepare samples to enable the execution of the ESD/HTOL test.

2) The QA/Design engineer prepares a test plan outlining how the ESD/HTOL Test will be conducted.
3) The QA engineer provides the ESD/RT test plan to a testing company such as QRT and requests the execution of the test.
   Refer to the follow test guide.
   Once the ESD test is completed by ESD test site, the QA engineer performs post measurement on the zapped samples.
4) After making the final judgment on the test results, the QA engineer prepares the test report.

Phase #4-1 ESD / HTOL Test guide

ESD Model	Test pin group	Ref. GND Pin group	Test level	Zap	Sample size	Judgment
HBM (Human Body Model)	VSS, IO	VDD	Target level (KV)	(+/-) 1	3	0/3
	VDD, IO	VSS, VSS_no_zap		(+/-) 1	3	0/3
	IO	IO		(+/-) 1	3	0/3
Note: 1st step _ Conduct an ESD limit test with a single sample each mode. 2nd step _ Conduct an ESD test with target level for qualification each mode						
CDM (Charged Device Model)	ALL Pin	N/A	Peak current 7A level	(+/-) 1	3	0/3
Note: 1st step _ To determine the for a peak current of 7A, CDM zapping test will be conducted. 2nd step _ Check for damage by measuring the pin I-V curve. 3rd step _ Conduct an CDM test with the found test level for qualification.						
Latch-up	Temperature condition: Room		100mA	Positive 1	3	0/3
				Negative 1	3	0/3
			Vmax *1.5	(+) 1	3	0/3

Reliability	Temperature	Stress voltage	Stress time	Sample size	Judgment
HTOL (High temperature operating life)	125 C	Operating Max V or the stress condition equivalent 10years operating life.	504hr / 1008hr	77	0/77

Phase #4-2 Guideline for judgment criteria

Test items	Input Voltage	0hr measurement	Post measurement	Criteria information

ESD_Latch-up_Guide.pptx

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Standby current (IDDQ)	Typ. V	0hr Current	Post current	**[Commercial IP or product]** De-gradation: x2 ↑ **[Automotive IP or product]** De-gradation: x1.5 ↑
Dynamic current (DIDD)	Min. V Max. V	0hr Current	Post current	**[Commercial IP or product]** De-gradation: 30% ↑ **[Automotive IP or product]** De-gradation: 20% ↑
Scan test	Min. V Max. V	Good die	Pass/fail under input voltage	Pass / Fail
Function test (External loopback test)	Min. V Max. V	Good die	Pass/fail under input voltage	Pass / Fail

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7. Relevant document

Material	Form code	Description
ESD_Latch-up_Guide	RS-MP-06-I-01	-. Explanation of ESD test Theory and test methods.

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